SECURING

AGENTIC AI SYSTEMS

An 8-Hour Hands-On Workshop

BSidesOK 2026 · Training Day · April 8

Evan Wagner · **Christopher Williams**

From ML Fundamentals to Agentic Defense

YOUR INSTRUCTORS

Meet the team behind the workshop

EVAN WAGNER

- Co-instructor & Workshop Co-Organizer
- Cybersecurity practitioner
- HackNWA, SFISSA, HackMiami
- Professional CSIRT Lead
- Experienced Programmer, DevOps, Ethical Hacker

CHRISTOPHER WILLIAMS

- Principal Incident Response Engineer
- Active CISSP · CISM · GCFA
- B.S. in Cyber Security from Bellevue University
- Graduate Student at Bellevue
- 30+ years industry experience
- Linux/Unix, DevOps, Ethical Hacker

My presentation, comments and opinions are provided in my personal capacity and not as a representative of my employer. They do not reflect the views of my employer and are not endorsed by my employer.

WORKSHOP REQUIREMENTS

What you need · All labs are browser-accessible

HARDWARE OPTIONS

- Linux with modern Nvidia GPU

— OR —

- Windows / Mac with hypervisor:

VMware Workstation / Fusion

VirtualBox

- Ollama-compatible system accepted

VM PROVIDED

- Preconfigured VM image:

.ova (VMware / VirtualBox)

.qcow2 (KVM / QEMU)

- All labs run in the browser
- No local install beyond VM host

TOOLS INCLUDED

Python · PyTorch · Transformers · Ollama · Open WebUI · Gradio · FastMCP · LangChain · LangGraph · OpenClaw · Elasticsearch · Wazuh

WORKSHOP AGENDA

8 hours · 10 labs · ML Fundamentals through Agentic Defense

Lab 1	ML Fundamentals	20 min
Lab 2	From ML to GenAI	20 min
Lab 3	LLMs Using Tools / MCP	45 min
— BREAK —		15-20 min
Lab 4	Building Agents	45 min
Lab 5	Autonomous Agents	45 min
Lab 6	Attacking LLMs	45 min
— LUNCH —		60 min
Lab 7	Watchword Observability	45 min
Lab 8	Detection & Alerting	45 min
— BREAK —		15-20 min
Lab 9	Canaries in the Coal Mine	45 min
Lab 10	IR Capstone	30 min
Q&A · Open Discussion		

LAB 1: ML FUNDAMENTALS

20 minutes · Classification · Learning Types · GLUE Benchmark

LECTURE

- Supervised · Unsupervised · Reinforcement
- Classification, regression, clustering
- Training / validation / test sets
- Overfitting, underfitting, bias-variance
- Model evaluation: accuracy, F1, AUC
- Neural networks & deep learning intro

LAB EXERCISE

- Label a small dataset manually
- Categories: Plant / Animal / Mineral
- Run the GLUE benchmark on a small purpose-built model
- How well does it classify?
- What factors affect accuracy?
- Tools: Python + Transformers

LAB 2: FROM ML TO GENAI

20 minutes · LLMs · Prompting · Consistency Testing

LECTURE

- Discriminative vs generative models
- What is an LLM? Tokens & attention
- Prompt engineering fundamentals
- Temperature & sampling parameters
- Local vs cloud-hosted models
- Ollama: running models locally

LAB EXERCISE

- Connect to a local LLM via Ollama
- Ask the same question multiple times
- How consistent are the responses?
- Use Open WebUI chat interface
- Can you tell you're chatting with AI?
- Try basic prompt manipulation
- Discuss: LLM behavior & limits

LAB 3: LLMs USING TOOLS · MCP PRIMER

45 minutes · Tool Calls · FastMCP · Server/Client Architecture

LECTURE

- What is a tool call?
- Model Context Protocol (MCP) overview
- MCP: host / client / server architecture
- Tool schemas and JSON-RPC transport
- How LLMs decide when to use tools
- FastMCP: Python MCP server builder
- Security implications of tool use

LAB EXERCISE

- Build a FastMCP server + client
- Expose: add, subtract, multiply, divide
- Watch the LLM invoke tool calls live
- Inspect the JSON-RPC messages
- Our agent can now compute!
- Discuss: What could go wrong?
- Preview: tool misuse attack surface

BREAK TIME

15 - 20 Minutes

LAB 4: BUILDING AGENTS

45 minutes · LangChain · ReAct Pattern · Agent Loops

LECTURE

- What is an AI agent?
- Agents vs chatbots vs pipelines
- The ReAct pattern: Reason + Act
- LangChain: chains, agents, runnables
- Agent components: LLM, tools, memory
- Execution loops and state management
- When do agents go wrong?

LAB EXERCISE

- Build an agent using LangChain
- Connect the tools from Lab 3
- Define a goal and watch it reason
- Observe Thought → Action → Observe
- Add a new tool and test behavior
- Inspect intermediate reasoning steps
- Discuss: autonomous decision risks

LAB 5: AUTONOMOUS AGENTS

45 minutes · OpenClaw · LangGraph · Multi-Step Autonomy

LECTURE

- Intro to OpenClaw framework
- LangGraph: stateful agentic graphs
- Multi-agent coordination patterns
- Goal decomposition and planning
- Persistence, checkpoints, memory
- Agentic attack surface expansion
- Real-world autonomous agent risks

LAB EXERCISE

- Launch OpenClaw with an objective
- Give the agent a multi-step mission
- Observe autonomous planning behavior
- What tools does it reach for?
- Try to redirect the agent mid-task
- Discuss containment strategies
- Why does this matter for security?

LAB 6: ATTACKING LLMs

45 minutes · 10 Attack Classes · Best Attack Wins Challenge Coins

Prompt Injection

Force unintended behavior via crafted input

Jailbreaking

Bypass safety guardrails and policies

Data Exfiltration

Extract training data or context window

Poisoning Attacks

Corrupt training or RAG data sources

Social Engineering

Manipulate users through AI output

Insecure Output

Exploit downstream processing of LLM output

Multimodal Attacks

Image or audio as attack vector

Tool / Agent Abuse

Invoke tools for unintended actions

Supply Chain

Compromise models, plugins, or MCP servers

DoS & Hybrid

Resource exhaustion and combined techniques

LUNCH BREAK

Fuel up · 4 labs remaining

LAB 7: WATCHWORD OBSERVABILITY

45 minutes · Logging · Elasticsearch · Evidence Collection

LECTURE

- "Logs or it didn't happen"
- What to log in agentic systems?
- Connection info & session metadata
- Prompts, responses, and tool calls
- Agent actions and decision traces
- Elasticsearch for log aggregation
- Structured vs freeform logging

LAB EXERCISE

- Review logs from the Lab 6 attacks
- Identify attack indicators in logs
- Use Elasticsearch to query events
- Find the data exfiltration attempt
- Correlate prompt + response + tool
- Build a timeline of malicious activity
- What would you have missed?

LAB 8: DETECTION AND ALERTING

45 minutes · Wazuh · Alert Rules · Needle in a Haystack

LECTURE

- Needle in a haystack: what matters?
- Signal vs noise in AI system logs
- Generating alerts from log events
- Wazuh: open-source SIEM / XDR
- Rule writing and alert correlation
- Severity tiering for AI-specific events
- MITRE ATT&CK for LLM-based threats

LAB EXERCISE

- Configure Wazuh alert rules
- Target: prompt injection signatures
- Target: abnormal tool call patterns
- Target: data exfil volume thresholds
- Trigger alerts using Lab 6 logs
- Tune rules to reduce false positives
- Discuss: alert fatigue vs blind spots

BREAK TIME

15 – 20 Minutes · Almost there!

LAB 9: CANARIES IN THE COAL MINE

45 minutes · Canary Tokens · Deception Defense · Detection

LECTURE

- What are canary tokens?
- Lure attackers · detect access
- Token types: URLs, files, credentials
- Planting tokens in AI contexts:
  - System prompts
  - Tool responses
  - RAG document stores
- LLM data exfil detection via canaries

LAB EXERCISE

- Plant canary tokens in the environment
- Embed in documents & tool outputs
- Trigger tokens via attack simulation
- Observe beacon callbacks in real time
- Discuss use cases:
  - RAG poisoning detection
  - LLM-based exfil detection
  - Insider threat identification

LAB 10: INCIDENT RESPONSE CAPSTONE

30 minutes · Attribution · Impact · Containment · Prevention

SCENARIO: A customer service AI agent has been compromised. Attackers targeted the public-facing application. You have alerts, traces, and logs. Find them.

MISSION

- Determine attribution
- Who carried out the attack?
- What data was obtained?
- How did they gain access?

METHOD

- Follow the alert trail
- Trace logs forward & backward
- Correlate tool call records
- Map to MITRE ATT&CK

OUTCOME

- Containment plan
- App hardening recommendations
- Detection rule improvements
- Prevention strategies

QUESTIONS?

Thank you for attending!

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All materials are open source. Labs, slides, and code will be shared.

SLIDE TITLE

Subtitle / context

[Content goes here]